

2.5. SUMMARY

Optimal asset allocation with labor income crucially depends on how risky your labor income is and how it correlates with equities. If your labor income is bond-like (lowly correlated with stock returns), treat it as such and hold more equities in your portfolio. Hold even more equities if your labor income acts the opposite way of equities. If your labor income is stock-like (highly correlated with stock returns), you will offset that equity risk by holding lots of bonds. Thus, paraphrasing Moshe Milevsky (2009), a pension finance expert at Canada's York University, *personal asset allocation depends on whether you are a stock or a bond*.

In summary, all else being equal:

Bond-like labor income $\Rightarrow$ Hold more equities.

Stock-like labor income $\Rightarrow$ Hold more bonds.

In real life, people seem to act as our theory predicts. Guiso, Jappelli, and Terlizzese (1996) and Vissing-Jørgensen (2002) show that individuals with very volatile wages tend not to hold equities, or, if they hold equities, they do so as a smaller share of their portfolio. Betermier et al. (2012) employ a data set that tracks 3% of the Swedish population and examine people changing jobs. As individuals move from jobs with less to more volatile wages, they reduce their exposure to risky assets. In their sample, the least volatile wages are earned by individuals working in the field of recycled metal waste and the most volatile wages are earned by those working in asset management (!). An individual moving from the former to the latter slashes her holdings of risky assets by 35%. Dimmock (2012) reports that universities with more volatile non-financial income (tuitions, fees, grants, donations, etc.) hold more fixed income and other low-risk securities. Very volatile nonfinancial income markedly reduces allocations to alternative assets, except for the most selective universities.

3. Life Cycle

Before we describe optimal asset allocation over the life cycle, let's review. . . .

3.1. FINANCIAL PLANNERS' ADVICE

Ubiquitous advice from financial professionals is to reduce your stock allocation as you age. A popular rule of thumb is *one hundred minus your age*, which says that the proportion of your portfolio invested in equities should equal the difference between your age and one hundred. There are many variants, but I will refer to any plan that reduces risky asset holdings as you approach retirement as the "one hundred minus your age rule."

Panel A of Table 5.1 lists the asset allocation for the Vanguard Target Retirement 2050 Fund. Such *target date funds* are designed to modify an investor's asset allocation as he approaches retirement—in this case, a date that falls between